Nba Three Point Prediction Blog Post

Predicting the NBA's Next Great Shooter: Can Machine Learning See a Breakout Coming?

Basketball has changed dramatically over the last twenty years. Teams now build entire offenses around the three-point shot, and players who can consistently shoot from deep have become some of the most valuable athletes in the league. But one question still fascinates fans, analysts, and front offices alike:

Can we predict who will become a better three-point shooter next season?

For our project, we built a machine learning model designed to predict how an NBA player's three-point percentage would change from one season to the next. Instead of simply guessing a player's future percentage outright, we focused on something more interesting: **improvement and decline**.

Would the model recognize when a player had a fluke shooting season? Could it identify players who were genuinely developing into elite shooters? And most importantly, could it outperform the simple assumption that "next year will look just like this year"?

The answer was yes, but with some important limitations.

The Core Idea

Most sports predictions are surprisingly conservative. If a player shoots 38% from three this season, the safest prediction for next season is usually… around 38%.

That simple strategy is called a **baseline model**, and it is much harder to beat than people think.

Our project attempted to improve on that baseline using machine learning. We trained models on NBA shot data from the 2004–2024 seasons and tried to predict changes in a player's shooting percentage based on factors such as:

- Shot location
- Shot volume
- Age

- Position
- Height and weight
- Shot distribution
- Previous shooting performance
- Whether the player was improving or declining over time

Instead of treating all three-pointers equally, we broke them into categories such as:

- Corner threes
- Above-the-break threes
- Left corner vs. right corner shooting

This allowed the model to understand *how* a player shoots, not just whether the shots went in.

The Dataset

We used publicly available NBA shot tracking data and player information.

Dataset Sources

- NBA shot data scraped from Kaggle and the NBA API
- Player bio information from the NBA API

You can explore the datasets here:

- Dataset link: https://www.kaggle.com/datasets/mexwell/nba-shots
- NBA API: urlnba_api documentationhttps://github.com/swar/nba_api

To make the project fully reproducible, all code, figures, and processed datasets are included in the Github repository.

Github Repository

- Github Repo: https://github.com/BenShonfeld/Predicting-the-NBA-s-Next-Great-Shooter

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Building the Features

One of the most important parts of any machine learning project is deciding what information the model should pay attention to.

A casual fan might only look at a player's overall three-point percentage. But basketball is more nuanced than that.

For example:

- Some players take mostly easy corner threes.
- Others take difficult off-the-dribble shots from far above the break.
- Some players shoot a high percentage on very low volume.
- Others maintain efficiency while taking ten attempts per game.

We wanted the model to capture these differences.

Shot Profile Matters

One particularly interesting metric we created measured how balanced a player's shot diet was.

Players who only shoot from one area of the floor can sometimes post misleading percentages. A role player who only takes wide-open corner threes may shoot 43%, but that does not necessarily mean they are an elite all-around shooter.

Meanwhile, players taking harder above-the-break threes may have lower percentages while still possessing stronger long-term shooting skill.

By separating shot zones and tracking attempt distribution, the model could better understand the difference between sustainable skill and temporary hot streaks.

The Models

We tested three approaches.

1. Baseline Model

The baseline simply predicted that a player's shooting percentage would not change.

This sounds simplistic, but it is actually a very strong benchmark because NBA shooting percentages tend to be relatively stable over time.

2. Linear Regression

Next, we trained a linear regression model.

This model attempts to identify relationships between variables in a relatively straightforward way. For example, it might learn that younger players tend to improve slightly, or that high-volume shooters regress toward the mean.

3. Artificial Neural Network (ANN)

Finally, we trained a small neural network.

Neural networks are better at capturing nonlinear relationships and more subtle interactions between variables. Instead of relying on one straight-line relationship, the ANN can combine many patterns together.

Because our dataset was relatively small compared to massive deep-learning projects, we intentionally kept the network lightweight.

The Results

The results were encouraging.

Mean Absolute Error (MAE)

Model	MAE
Baseline (No Change)	0.0435
Linear Regression	0.0359
Neural Network	0.0361

Root Mean Squared Error (RMSE)

Model	RMSE
Baseline (No Change)	0.0577
Linear Regression	0.0477
Neural Network	0.0481

Both machine learning approaches outperformed the baseline model.

Interestingly, the neural network did **not** dramatically outperform linear regression. This suggests that many of the relationships in the data may actually be fairly structured and interpretable rather than extremely chaotic.

In other words: basketball shooting improvement may be more predictable than people think.

Did the Model Understand Basketball?

One of the most interesting parts of this project was looking beyond the error metrics and asking:

Did the model actually learn meaningful basketball concepts?

This ended up being more important than the raw numbers themselves.

Catching Shooting Outliers

The model was fairly good at identifying players who had unsustainably hot shooting seasons.

For example, role players who suddenly shot far above their career average on low volume often got predicted to decline the following season.

This is a classic example of **regression toward the mean**.

A player shooting 47% from three on limited attempts is probably not secretly the greatest shooter in NBA history. More often, randomness, shot quality, lineup context, or temporary confidence spikes are involved.

Our models — especially the linear regression model — were surprisingly effective at recognizing these cases.

Recognizing Sustainable Improvement

The harder challenge was identifying players who genuinely improved.

This is difficult because real skill growth is not always obvious immediately. Young players may suddenly improve their mechanics, confidence, shot selection, or role within an offense.

The model sometimes caught these trends early when several indicators aligned:

- Increased shooting volume
- Better balance across shot zones
- Gradual year-over-year efficiency improvement
- Stable shooting mechanics reflected in consistency across shot locations

However, the model still struggled with massive leap seasons.

Players who completely reinvented themselves offensively were difficult to predict because those transformations often depend on factors the dataset could not measure:

- Coaching changes
- Mechanical adjustments
- Increased playing time
- Mental confidence
- Team offensive systems
- Injury recovery

In many ways, these “breakout” players remain one of the hardest prediction problems in sports analytics.

Why the Neural Network Wasn’t Much Better

One surprising result was how close the neural network and linear regression models ended up.

Normally, people assume neural networks automatically dominate traditional models.

But in this case, the simpler model performed almost identically.

There are several possible reasons:

1. NBA shooting percentages are already relatively stable.
2. The dataset was not enormous.
3. Many of the important relationships were fairly linear.
4. Human shooting development may simply contain a lot of randomness.

This became an important lesson in itself.

Machine learning is powerful, but more complexity does not always mean better predictions.

Sometimes a well-designed simple model can compete surprisingly well.

What We Learned

This project taught us that sports analytics is about much more than simply building a model.

The real challenge is deciding:

- What information matters?
- Which patterns are sustainable?
- What separates luck from skill?
- How do you measure development?

Basketball statistics contain both signal and noise.

Our goal was not to create a perfect predictor of the future. Instead, we wanted to build a system that could recognize meaningful patterns in player shooting behavior.

In many cases, it succeeded.

The model consistently outperformed the assumption that players simply stay the same from year to year. It also showed signs of understanding important basketball concepts like shot quality, volume, and regression toward the mean.

At the same time, it struggled with the exact thing humans struggle with too:

predicting when a player will suddenly become much better.

And honestly, that may be part of what makes sports so compelling.

Future Improvements

If we continued this project, there are several areas we would want to explore:

Tracking Shot Difficulty

Not all three-point attempts are equally hard. Including defender distance, shot clock pressure, or pull-up vs. catch-and-shoot data could improve predictions.

Incorporating Playstyle Data

A player's role within an offense matters enormously. Some players create their own shots while others rely entirely on teammates.

Sequential Modeling

Instead of treating each season independently, future versions could use sequence-based models that better capture long-term player development.

More Contextual Variables

Coaching systems, teammate spacing, injuries, and minutes played could all add useful information.

Final Thoughts

Sports fans often debate whether shooting improvement is “real” or just luck.

Our project suggests the answer is somewhere in the middle.

Some changes are predictable. Others are not.

Machine learning can identify broad patterns and even recognize certain types of sustainable improvement, but basketball still contains enough unpredictability to surprise both fans and algorithms alike.

And maybe that is exactly how it should be.

Repository and Reproducibility

All code, visualizations, preprocessing scripts, and datasets used in this project are available in the Github repository below.

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- Dataset Link: https://www.kaggle.com/datasets/mexwell/nba-shots